Reverse Linked List

AlgoExpert · Hard · Linked Lists

The problem

- Reverse in place and return the new head.

Intuition

- Flip each link to point backwards, keeping three pointers so nothing is lost.

Brute force vs optimal

- Copy into array, rebuild → O(n) space
- Recursive → O(n) stack
- Iterative three pointers → O(1)

Approach

- 1. Previous starts null
- 2. Save next, point current at previous
- 3. Advance both
- 4. Previous is the new head

Worked example

reversed (prev)



remaining (head)



After 0 steps

Worked example

reversed (prev)



remaining (head)



After 2 steps

Worked example

reversed (prev)



remaining (head)



After 4 steps

Worked example

reversed (prev)



remaining (head)



After 6 steps

Java solution

```java
public class ReverseLinkedList {
    public static LinkedListNode reverseLinkedList(LinkedListNode head) {
        LinkedListNode prev = null;
        while (head != null) { LinkedListNode n = head.next; head.next = prev; prev = head; head = n; }
        return prev;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Save next before overwriting
- Empty / single node
- Return prev, not head

Interview tips & follow-ups

- Follow-up: reverse in groups of k

Recap

- Flip each link to point backwards, keeping three pointers so nothing is lost.
- Complexity: O(n)